

Fourier analysis – 2

Fourier coefficients and smoothness

Homework

13 сентября 2026 года

Theoretical minimum. Differentiation multiplies the coefficients by in , so smoothness and decay are the same statement read in two directions, and Parseval makes the correspondence exact in the mean square. A Hölder condition of order α gives $|c_n| = O(n^{-\alpha})$ by the shift $x \mapsto x + \pi/n$, and for $\alpha > \frac{1}{2}$ Bernstein's theorem upgrades this to absolute convergence. In the other direction, coefficients of an L^1 function tend to zero but can do so as slowly as one likes, and not every null sequence is a coefficient sequence — the test $\sum b_n/n < \infty$ for odd coefficients is the cheapest obstruction. Structural hypotheses on the coefficients — one-sided support, convexity, lacunarity, finiteness — translate into strong structural statements about the function.

1. Prove that the exponent $\frac{1}{2}$ in Bernstein's theorem cannot be lowered: construct a function Hölder of order $\frac{1}{2}$ whose Fourier series does not converge absolutely.
2. (Капачёв 1.76) Let $(a_n)_{n \geq 0}$ be positive, decreasing to zero and convex, that is $a_{n-1} - 2a_n + a_{n+1} \geq 0$ for $n \geq 1$. Prove that the sum $\frac{a_0}{2} + \sum_{n \geq 1} a_n \cos nx$ is non-negative at every point where it converges.
3. (Капачёв 1.77) Prove that $f(x) = \sum_{n \geq 1} 2^{-n} \cos(2^n x)$ is nowhere differentiable. (Note that the theorem of Differentiable functions – 8 does not apply: there the hypothesis is $ab > 1 + \frac{3\pi}{2}$, while here $ab = 1$.)
4. Prove the Riemann–Lebesgue lemma for L^1 functions, and construct an integrable function whose Fourier coefficients tend to zero more slowly than any prescribed sequence $\varepsilon_n \rightarrow 0$.

5. Prove that if (b_n) are the sine coefficients of an integrable function, then $\sum \frac{b_n}{n}$ converges; deduce that $\sum \frac{\sin nx}{\ln n}$ is not the Fourier series of any integrable function, although it converges everywhere.
6. Let f be real-valued and integrable with $c_n(f) = 0$ for all $n < 0$. Prove that f is constant; and describe all complex-valued f with this property.
7. Prove that the Wiener algebra $A(\mathbb{T}) = \{f : \sum |c_n| < \infty\}$ is a Banach algebra under pointwise multiplication, and state Wiener's lemma about $1/f$.
8. Prove Jackson's inequality $E_n(f) \leq C\omega(f, \frac{1}{n})$ for the best uniform approximation by trigonometric polynomials, and its converse of Bernstein type: if $E_n(f) = O(n^{-\alpha})$ with $0 < \alpha < 1$, then f is Hölder of order α . (The inequality $\|T'\| \leq n\|T\|$ is proved in Trig-6.)
9. Compute the Fourier coefficients of the Fejér and Poisson kernels, and explain in terms of them why both families are approximate identities while the Dirichlet kernel is not.
10. Prove Wirtinger's inequality by Fourier series: if f is 2π -periodic, C^1 and $\int_0^{2\pi} f = 0$, then $\int_0^{2\pi} f^2 \leq \int_0^{2\pi} (f')^2$ with equality only for $f = a \cos x + b \sin x$; compare with the proof through the derivative in Differentiable functions – 9.